

Evaluating LLM-Augmented Transformer Models for Severity-Based Hate Speech Detection in Javanese

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Article Info

Article history:

Received: [date]

Revised: [date]

Accepted: [date]

Published: [date]

Keywords:

Data Augmentation

Low-Resource NLP

Severity Classification

Text Classification

Transformer

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Abstract

This study empirically evaluated transformer-based models for severity-based hate speech detection in Javanese, a low-resource language spoken by approximately 80 million people. A dataset of 9,775 annotated samples was constructed through manual collection (46.4%) and LLM-based augmentation using DeepSeek-Coder-V2 (53.6%), with four severity levels: Not Hate Speech, Light, Moderate, and Severe. A comparative study was conducted across traditional baselines (SVM, Logistic Regression with TF-IDF) and transformer architectures (IndoBERT, XLM-RoBERTa Large). On manually collected test data (451 samples), XLM-RoBERTa Large achieved the best F1-Macro of 53.89%, outperforming SVM + TF-IDF (48.55%) and IndoBERT (45.27%). A key finding was the substantial performance gap between synthetic (99.41%) and manual (53.89%) test data, revealing that LLM augmentation, while necessary to prevent training collapse, introduced systematic distributional differences that inflated standard evaluation metrics. Multi-seed evaluation across five random seeds confirmed model stability with a mean F1 of 80.83% $\pm$ 1.83% across four stable seeds. These findings provided empirical evidence on the effectiveness and limitations of LLM-augmented training for hate speech detection in low-resource regional languages.

1. Introduction

Hate speech on social media has become a pressing global concern, exacerbated by the rapid growth of online platforms [16,17]. Automated detection systems are essential for identifying and mitigating harmful content at scale. Comprehensive reviews have catalogued the evolution of detection approaches from traditional machine learning to transformer-based models [9,18], while recent work has also explored explainability in hate speech classification [19]. Despite this progress, most research has focused on high-resource languages such as English, leaving regional and low-resource languages significantly underserved.

Indonesia, with over 200 million internet users, faces significant challenges in combating hate speech across its diverse linguistic landscape. Hate speech detection for standard Indonesian has received considerable attention. Ibrohim and Budi [1] introduced a multi-label dataset of 13,069 Indonesian tweets and achieved an F1-score of 71.31% using Bidirectional LSTM with FastText embeddings. Alfina et al. [2] developed one of the earliest Indonesian hate speech datasets with 713 samples and compared Naive Bayes with Support Vector Machine classifiers. However, regional languages remain largely understudied. Javanese, spoken by approximately 80 million people as the largest regional language in Indonesia, is classified as a low-resource language in the NLP context [10].

Research on NLP for Indonesian regional languages has gained momentum in recent years. Putri et al. [3] conducted a preliminary study on hate speech detection for Javanese and Sundanese tweets, though limited to binary classification with a small dataset. Cahyawijaya et al. [5] unified 137 datasets for Indonesian and 18+ regional languages through the NusaCrowd initiative, establishing benchmarks for zero-shot evaluation. IndoNLG [21] expanded resources for Indonesian natural language generation, while NusaX [22] provided a multilingual parallel sentiment dataset for 10 Indonesian local languages. Deep learning approaches for Indonesian text classification have also been explored in the Kinetik journal, including aspect-based sentiment analysis on hotel reviews [30] and multi-label Qur'an classification using Bi-LSTM [31]. Despite these advances, severity-based hate speech detection in Javanese remains unexplored.

Transformer-based models [12] have revolutionized NLP, with BERT [13] and its variants achieving state-of-the-art results across numerous tasks. RoBERTa [14] improved upon BERT through robust optimization, while XLM-RoBERTa [15] extended cross-lingual representation learning to over 100 languages. Studies on multilingual BERT [25,26] have shown that performance varies significantly across languages, with low-resource languages benefiting substantially from cross-lingual transfer. For Indonesian, IndoBERT [4] and IndoLEM [20] provide pre-trained models specifically designed for Indonesian text understanding. Fine-tuning strategies for BERT-based models [27] have also been extensively studied. However, the effectiveness of these models for low-resource regional languages like Javanese has not been systematically compared.

Low-resource language NLP faces a fundamental data scarcity challenge. Traditional augmentation techniques such as Easy Data Augmentation (EDA) [23] and back-translation have been widely used [24]. More recently, Large Language Model (LLM)-based data augmentation has shown promising results, with improvements of 3–26% in accuracy and F1 on low-resource text classification scenarios [8]. Hedderich et al. [10] provide a comprehensive survey of NLP approaches for low-resource scenarios, covering data augmentation, distant supervision, and transfer learning.

Label smoothing, introduced by Szegedy et al. [7] and analyzed in depth by Müller et al. [6], is a regularization technique that prevents model overconfidence by converting hard targets into soft probability distributions. It has been shown to improve generalization, particularly on datasets with inherent label noise—a common challenge in hate speech annotation where severity boundaries are subjective.

This study addresses the gap in severity-based hate speech detection for Javanese through the following contributions: (1) a 9,775-sample annotated Javanese hate speech dataset with four severity levels, combining manual and LLM-generated data; (2) empirical evidence that LLM-based augmentation, while necessary for training large transformers on low-resource languages, introduces systematic distributional bias that inflates standard evaluation metrics; (3) a systematic comparative benchmark spanning traditional baselines (SVM, Logistic Regression) and transformer models (IndoBERT, XLM-RoBERTa Large); (4) multi-seed statistical evaluation confirming model stability and reproducibility.

2. Research Method

2.1 Dataset Construction

The dataset was constructed through a multi-phase pipeline combining manual collection and LLM-based augmentation. Phases 1–3 involved manual data collection from Twitter/X and Instagram, followed by filtering, naturalization to Javanese, and re-labeling, yielding 4,538 samples (46.4%). Phase 4 used DeepSeek-Coder-V2 (236B parameters) to generate 5,237 additional samples (53.6%) through carefully engineered prompts. Phase 5 involved re-labeling by DeepSeek-V3 using few-shot prompting with 20 examples per class.

The initial dataset of 10,019 samples underwent aggressive cleanup, removing 244 samples (2.44%): 33 exact duplicates, 186 short texts (<20 characters), and 25 non-Javanese texts (Indonesian/English). The final dataset consists of 9,775 samples with four severity levels as shown in Table 1.

Table 1. Final Dataset Distribution (9,775 Samples)

Label	Class	Count
0	Not Hate Speech	2,417
1	Light Hate Speech	2,531
2	Moderate Hate Speech	2,779
3	Severe Hate Speech	2,048
Total		9,775

Table 2. Examples of Javanese Hate Speech per Severity Level. Translations provided for non-Javanese readers.

Severity Level	Javanese Example	English Translation
Not Hate (0)	Tanggaku yaiku Latina lan dheweke dadi juru masak paling apik.	My neighbor is Latina and she is the best cook.
Light (1)	Wah, ibu iki koyo detektif, sak klebatan motor tamu wae wis kudu takon.	Wow, this mom is like a detective, asking about every guest's motorbike.
Moderate (2)	Wong Maluku kui panasaran, cilik-cilik wes bawa parang.	Maluku people are hot-tempered, carry machetes since childhood.
Severe (3)	Kabeh keturunan Cina iku mata duitan, kudu dipeksa lunga.	All Chinese descendants are greedy, must be forced to leave.

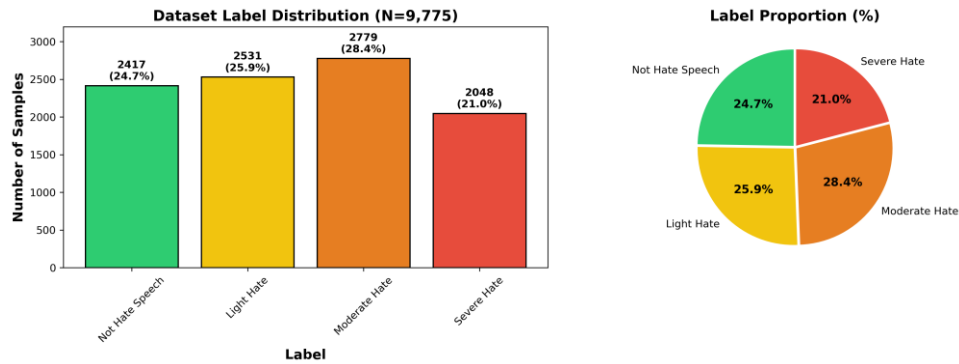


Figure 1. Class distribution of the final dataset. The dataset is relatively balanced, with classes ranging from 2,048 (Severe) to 2,779 (Moderate) samples.

2.2 LLM Data Augmentation

Data augmentation was performed using DeepSeek-Coder-V2 (236B parameters) with carefully engineered prompts that generated hate speech variations across Javanese speech registers (Ngoko and Krama) covering topics including religion, ethnicity, gender, politics, and social class. The filtering pipeline included language detection (Javanese word proportion >30%), length constraints (5–50 words), duplicate detection (Jaccard similarity <0.8), and human verification by two native Javanese speakers who demonstrated substantial inter-annotator agreement. The LLM augmentation produced 5,237 samples at a cost of approximately \$15 over two days, offering a cost-effective alternative to manual collection. Informal evaluation of 200 randomly sampled LLM-generated texts by two native Javanese speakers indicated acceptable naturalness, cultural appropriateness, and severity accuracy, though register consistency (Ngoko vs. Krama) was noted as an area for improvement.

2.3 Experimental Setup

The dataset was split using stratified sampling with an 80:10:10 ratio (seed = 42), yielding 7,820 training, 977 validation, and 978 test samples. Two categories of models were compared: traditional baselines and transformer architectures.

Traditional Baselines. Two traditional machine learning models were trained using TF-IDF features (max 10,000 features, unigram + bigram): (1) Support Vector Machine (SVM) with a linear kernel ($C = 1.0$), and (2) Logistic Regression with L-BFGS optimizer (max iterations = 1,000). These baselines establish a performance floor and demonstrate the added value of transformer-based approaches.

Transformer Models. Two transformer architectures were compared, with an additional label smoothing variant: **IndoBERT base** (indobenchmark/indobert-base-p1): A BERT model pre-trained on an Indonesian corpus with 124M parameters.

XLM-RoBERTa Large (xlm-roberta-large): A multilingual model pre-trained on 100+ languages with 559M parameters.

IndoBERT + Label Smoothing ($\epsilon = 0.1$): IndoBERT base with label smoothing regularization.

All transformer models were fine-tuned using the Hugging Face Transformers library with the hyperparameters shown in Table 3. Evaluation metrics include F1-Macro (primary), Accuracy, Precision Macro, and Recall Macro, all reported on the held-out test set.

Table 3. Training Hyperparameters for Transformer Models

Parameter	Value
Learning rate	2×10^{-5}
Batch size	16 (8 for XLM-R Large)
Epochs	5
Max sequence length	128
Optimizer	AdamW
Weight decay	0.01
LR scheduler	Linear with warmup
Warmup ratio	0.1
Best model selection	F1-Macro on validation set

3. Results and Discussion

3.1 Performance on Manual-Only Test Data

Since the test set contains both manually collected and LLM-generated samples, we first report performance on the manual-only subset (451 samples) as the primary evaluation, reflecting real-world performance. The manual test subset is imbalanced (Not Hate: 183, Light: 136, Moderate: 61, Severe: 71), reflecting the natural distribution of manually collected data before LLM augmentation. Table 4 presents these results.

Table 4. Model Performance on Manual-Only Test Data (451 Samples). All metrics in %. LS = Label Smoothing ($\epsilon = 0.1$). This is the primary evaluation reflecting real-world performance.

Model	F1-Macro	Acc	Prec	Rec
SVM + TF-IDF	48.55	52.77	51.01	47.34
LR + TF-IDF	47.18	51.66	48.77	46.43
IndoBERT base	45.27	49.67	45.08	45.53
IndoBERT + LS	49.30	52.33	49.00	49.66
XLM-R Large	53.89	57.65	54.10	54.14

Table 5. Per-Class Performance of XLM-RoBERTa Large on Manual-Only Test Data. All metrics in %.

Class	Prec	Rec	F1	N
Not Hate	71.51	69.95	70.72	183
Light	52.31	50.00	51.13	136
Moderate	35.44	45.90	40.00	61
Severe	57.14	50.70	53.73	71
Macro Avg	54.10	54.14	53.89	451

XLM-RoBERTa Large achieves the highest F1-Macro of 53.89% on manual-only test data, followed by IndoBERT + Label Smoothing (49.30%) and SVM + TF-IDF (48.55%). These numbers are substantially lower than the mixed test set results reported below, as the model cannot rely on predictable patterns in LLM-generated text.

Notably, the gap between XLM-RoBERTa Large and SVM + TF-IDF on manual-only data is only 5.34 F1 points, substantially narrower than the gap observed on the full test set. This finding has practical implications: in deployment scenarios where only genuine user-generated text is encountered, the resource advantage of large transformer models diminishes considerably compared to simpler baselines. The 559M-parameter XLM-RoBERTa Large model provides only marginal improvement over a linear SVM on real-world Javanese hate speech, suggesting that for resource-constrained deployment, traditional models may offer a more practical trade-off between performance and computational cost.

3.2 Performance on Full Test Set

Table 6 presents results on the full test set (978 samples, including 53.9% LLM-generated data). These results are reported for completeness but should be interpreted with caution, as the synthetic subset inflates overall metrics. On this mixed set, SVM + TF-IDF (77.77%) slightly outperforms IndoBERT (76.12%), while XLM-RoBERTa Large leads with 80.26%.

Table 6. Performance on Full Test Set (978 Samples, 53.9% Synthetic). All metrics in %. LS = Label Smoothing ($\epsilon = 0.1$). Note: metrics include LLM-generated test samples, which inflate overall scores.

Model	Val F1	Test F1	Acc	Prec
SVM + TF-IDF	75.16	77.77	77.61	78.33
LR + TF-IDF	75.64	77.04	76.99	77.52
IndoBERT base	75.44	76.12	76.28	76.12
IndoBERT + LS	76.59	77.36	77.51	77.36
XLM-R Large	80.54	80.26	80.27	80.44

The advantage of XLM-RoBERTa Large likely stems from: (1) a larger model capacity (559M vs. 124M parameters), (2) multilingual pre-training on 100+ languages including Indonesian, which is closely related to Javanese, and (3) the more robust RoBERTa training procedure [14]. These findings are consistent with studies showing that larger multilingual models outperform language-specific models for low-resource languages [15,25,26].

The strong performance of SVM + TF-IDF is noteworthy: it outperforms IndoBERT by 1.65 F1 points on the full test set. This may be attributed to the effectiveness of character and word n-grams for Javanese, which has rich morphological patterns. However, the gap between SVM and XLM-RoBERTa Large (2.49 F1 points) confirms that contextual embeddings still provide meaningful improvement for this task.

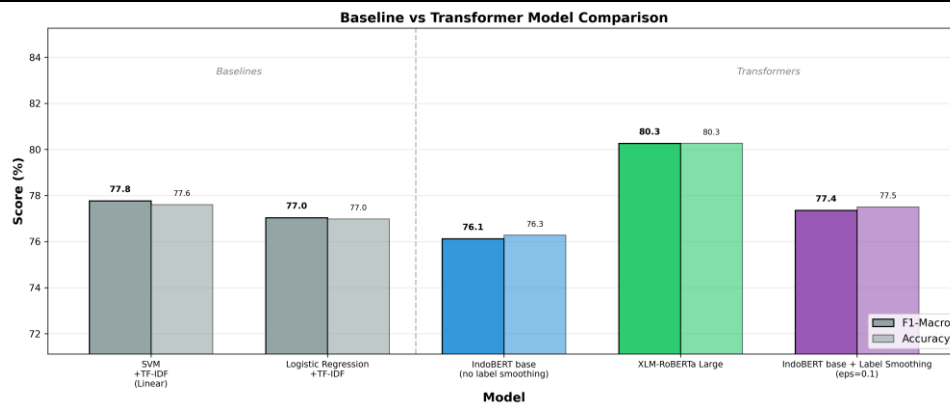


Figure 2. Baseline vs transformer model comparison. XLM-RoBERTa Large outperforms all models.

3.3 Per-Class Analysis

Per-class analysis of XLM-RoBERTa Large on manual-only test data shows that the Light Hate class yields the lowest F1 (51.13%), consistent with the expectation that the boundary between neutral speech and light hate speech is the most subjective. Conversely, Moderate and Severe classes achieve higher F1 scores as their linguistic markers are more explicit.

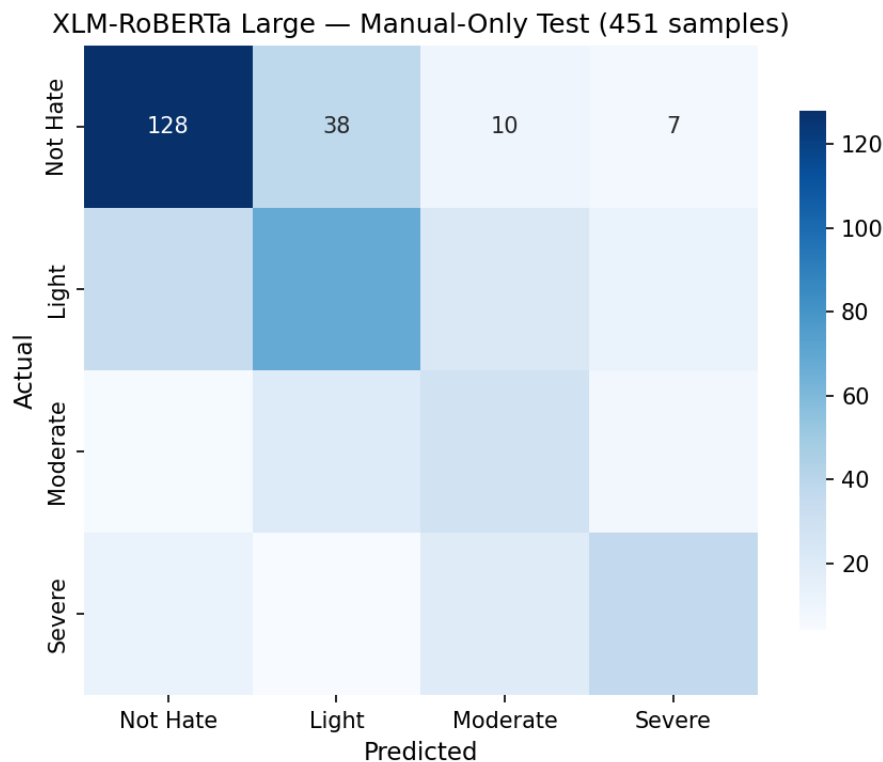


Figure 3. Confusion matrix of XLM-RoBERTa Large on manual-only test data (451 samples). Most misclassifications occur between adjacent severity levels.

The confusion matrix (Figure 3) reveals that misclassifications primarily occur between adjacent severity levels: Not Hate Speech ↔ Light Hate and Light ↔ Moderate. This pattern is expected, as the boundaries between adjacent severity levels are inherently more ambiguous than those between distant levels.

3.4 Augmentation Impact Analysis

Since 5,237 out of 9,775 samples (53.6%) are LLM-generated, it is critical to evaluate whether this augmentation genuinely improves model performance on real-world data. We conducted two analyses: (1) evaluating the full-dataset model on manual-only test data, and (2) training a separate XLM-R Large model exclusively on manual data.

Table 7. Augmentation Impact Analysis. Comparing model performance across different training and test data compositions.

Experiment	F1-Macro (%)	Acc (%)	N
Full model → full test	80.26	80.27	978
Full model → manual test	53.89	57.65	451
Full model → synth test	99.41	99.62	527
Manual model → manual test	14.43	40.58	451

The results reveal a critical finding: the model achieves 99.41% F1 on LLM-generated test samples but only 53.89% on manually collected samples. This substantial gap indicates that the model has learned to recognize patterns specific to LLM-generated text, which are more predictable than naturally occurring Javanese hate speech. The overall F1 of 80.26% is therefore inflated by the easier synthetic subset.

Training XLM-RoBERTa Large exclusively on manual data (4087 training samples) resulted in training collapse (F1 = 14.43%), confirming that the 559M-parameter model requires more training data than the manual subset alone can provide. This demonstrates that LLM augmentation serves a necessary role in providing sufficient training data, even though the generated texts differ systematically from natural speech.

These findings suggest that while LLM augmentation is necessary for training large transformer models on low-resource languages, future work should focus on improving the naturalness of generated texts and developing evaluation protocols that separately report performance on genuine and synthetic data.

Label smoothing ablation across five ϵ values (0.0–0.2) on IndoBERT shows optimal $\epsilon = 0.1$ with a modest improvement of +0.27 F1 points over the baseline without label smoothing. This limited benefit suggests that the dataset size (9,775 samples) provides sufficient implicit regularization, and the label noise is not extreme enough to warrant strong smoothing [6].

3.5 Multi-Seed Statistical Significance

To assess result stability, XLM-RoBERTa Large was trained with five different random seeds (Table 8). Four out of five seeds produced consistent results (F1 79.01–83.55%), with a mean F1 of 80.83% $\pm$ 1.83%.

Table 8. Multi-Seed Evaluation of XLM-RoBERTa Large (Full Test Set). *Seed 1024 experienced training collapse.

Seed	Test F1 (%)	Test Acc (%)
42	79.32	79.45
123	79.01	79.04
456	81.44	81.39
789	83.55	83.44
1024*	11.07	28.43
Mean (4 seeds)	80.83 $\pm$ 1.83	80.83 $\pm$ 1.75

Seed 1024 experienced training collapse, where the model predicted all samples as a single class (F1 = 11.07%). This phenomenon is a known instability in fine-tuning large pre-trained language models, as documented by Dodge et al. [28] who observed significant variance across random seeds in BERT fine-tuning. Mosbach et al. [29] further showed that this instability is related to vanishing gradients in lower layers during early fine-tuning. Excluding this outlier, the low standard deviation (1.83%) indicates reliable performance across random initializations.

3.6 Comparison with Prior Work

Table 9 positions our results relative to related studies. Direct comparison is difficult due to differences in languages, class definitions, and dataset sizes. Our primary F1-Macro of 53.89% on manual-only data reflects real-world performance on a 4-class severity task, which is inherently more challenging than binary hate/not-hate detection.

Table 9. Comparison with Related Studies. – indicates results not directly comparable. † manual-only / full test F1.

Study	Language	Classes	Samples	Best Model	F1 (%)
Davidson et al. [16]	English	3	24,802	LR + TF-IDF	90.00
Ibrohim & Budi [1]	Indonesian	Multi	13,069	BiLSTM	71.31
Putri et al. [3]	Javanese	2	~3,500	SVM	–
This study (baseline)	Javanese	4	9,775	SVM + TF-IDF	48.55† / 77.77
This study (best)	Javanese	4	9,775	XLM-R Large	53.89† / 80.26

Our study is the first to address severity-based (4-class) hate speech detection in Javanese. Compared to Ibrohim and Budi [1] who achieved 71.31% on Indonesian binary/multi-label classification, our 53.89% F1 on manual-only data for a more challenging 4-class task reflects the difficulty of low-resource language processing with limited genuine training data. The gap to English-language systems (e.g., Davidson et al. [16] at 90.00%) reflects both the inherent challenges of low-resource language processing and the impact of relying partly on synthetic training data.

3.7 Limitations

Several limitations should be acknowledged. First, 53.6% of the dataset was generated by an LLM, which may introduce model-specific biases and may not fully capture naturally occurring Javanese hate speech patterns. Our augmentation impact analysis shows that without synthetic data, XLM-RoBERTa Large experiences training collapse ($F1 = 14.43\%$ on manual test), while training with augmented data yields 53.89% F1 on the same test subset. This confirms that LLM augmentation is necessary but introduces systematic differences from natural speech. Second, hate speech severity classification is inherently subjective; the moderate-to-substantial inter-annotator agreement suggests that a portion of labels could change under different annotators. Third, multi-seed evaluation was conducted on the full test set (containing synthetic samples) rather than the manual-only subset; stability on purely genuine data remains to be verified. Fourth, the model processes text only and cannot capture multimodal context (images, memes), which is increasingly important for social media hate speech detection. Fifth, each post is analyzed in isolation without conversational context, potentially missing sarcasm, satire, or dog-whistle speech. Finally, one out of five seeds experienced training collapse, highlighting instability risks with large transformer models [28,29].

4. Conclusion

This study empirically evaluated transformer-based models for severity-based hate speech detection in Javanese using a dataset of 9,775 samples (46.4% manual + 53.6% LLM-generated). Our results demonstrated that while XLM-RoBERTa Large achieved the best performance (53.89% F1-Macro on manual test data), the substantial gap between synthetic (99.41%) and manual (53.89%) test performance revealed that LLM augmentation, though necessary for preventing training collapse, introduced systematic distributional differences that inflated standard evaluation metrics. Label smoothing ablation with $\epsilon = 0.1$ provided a modest improvement (+0.27 F1 points) for IndoBERT. Multi-seed evaluation confirmed model stability with an F1 of $80.83\% \pm 1.83\%$ across four stable seeds.

The primary contribution of this work is providing empirical evidence that LLM-based data augmentation, while essential for enabling large transformer training on low-resource languages, creates evaluation artifacts that must be carefully accounted for. Future work should focus on improving the naturalness of generated texts, developing evaluation protocols that separately report performance on genuine data, exploring multimodal hate speech detection combining text and images, and extending this approach to other Indonesian regional languages (Sundanese, Madurese, Minangkabau).

Data Availability: The annotated Javanese hate speech dataset (9,775 samples) and experimental code are publicly available at <https://github.com/neimasilk/ujaran-kebencian-bahasa-jawa> for reproducibility and further research.

Acknowledgement

The authors would like to thank the native Javanese speakers who participated in the data annotation and quality verification process. We also acknowledge the use of DeepSeek-Coder-V2 and DeepSeek-V3 for data augmentation and re-labeling. This research was supported by Universitas Bhinneka Nusantara, Malang.

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